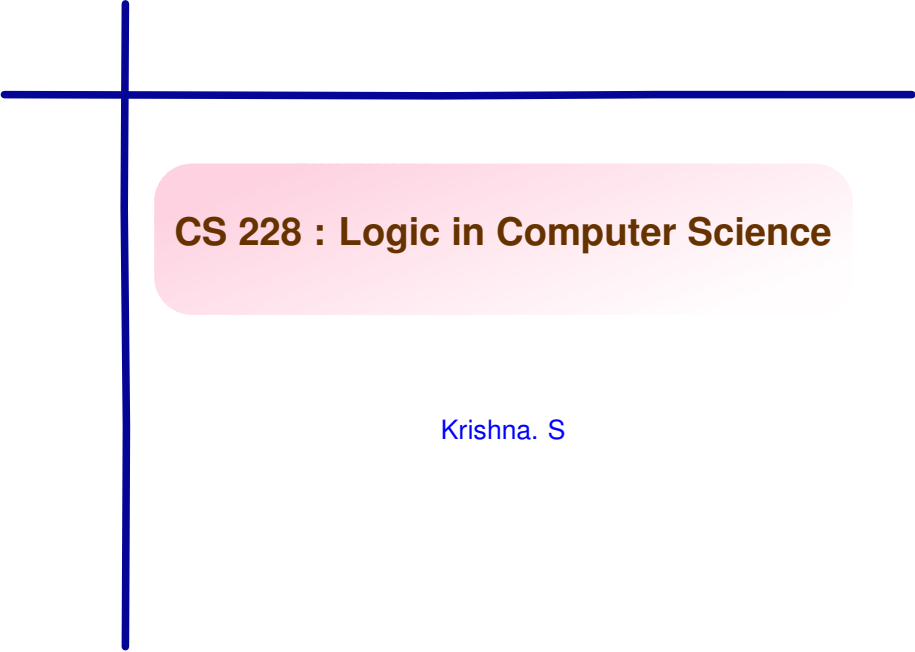


A decorative blue crosshair consisting of a vertical line and a horizontal line intersecting in the upper-left quadrant of the slide.

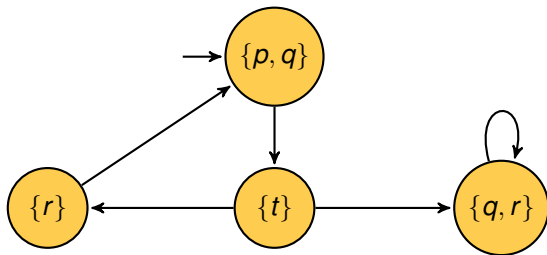
CS 228 : Logic in Computer Science

Krishna. S

Recap

- ▶ Transition Systems
- ▶ Traces

Specifying Traces



- ▶ Whenever p is true, r will eventually become true
 - ▶ $\{A_0A_1A_2\cdots \mid \forall i \geq 0, p \in A_i \rightarrow \exists j \geq i, r \in A_j\}$
- ▶ q is true infinitely often
 - ▶ $\{A_0A_1A_2\cdots \mid \forall i \geq 0, \exists j \geq i, q \in A_j\}$
- ▶ Whenever r is true, so is q
 - ▶ $\{A_0A_1\cdots \mid \forall i \geq 0, r \in A_i \rightarrow q \in A_i\}$

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Given AP , a set of propositions,

- ▶ Propositional logic formulae over AP
 - ▶ $a \in AP$ (atomic propositions)
 - ▶ $\neg\varphi, \varphi \wedge \psi, \varphi \vee \psi$
- ▶ Temporal Operators
 - ▶ $\bigcirc\varphi$ (Next φ)
 - ▶ $\varphi \mathbf{U}\psi$ (φ holds until a ψ -state is reached)
- ▶ LTL : Logic for describing LT properties

Semantics (On the board)

LTL formulae φ over AP interpreted over words $w \in \Sigma^\omega$, $\Sigma = 2^{AP}$,
 $w \models \varphi$

Derived Operators

- ▶ $true = \varphi \vee \neg\varphi$
- ▶ $false = \neg true$
- ▶ $\diamond\varphi = true \text{ U } \varphi$ (Eventually φ)
- ▶ $\Box\varphi = \neg\diamond\neg\varphi$ (Forever φ)

Precedence

- ▶ Unary Operators bind stronger than Binary
- ▶ $\bigcirc$ and $\neg$ equally strong
- ▶ U takes precedence over $\wedge, \vee, \rightarrow$
 - ▶ $a \vee b \text{ U } c \equiv a \vee (b \text{ U } c)$
 - ▶ $\bigcirc a \text{ U } \neg b \equiv (\bigcirc a) \text{ U } (\neg b)$

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 $\Box(\text{red} \rightarrow \neg \bigcirc \text{green})$
- ▶ Eventually the traffic light will become yellow
 $\Diamond \text{yellow}$
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 $\Box(\text{yellow} \rightarrow \Diamond \text{green})$

Semantics over Infinite Words

Given LTL formula φ over AP ,

$$L(\varphi) = \{\sigma \in (2^{AP})^\omega \mid \sigma \models \varphi\}$$

Let $\sigma = A_0A_1A_2\dots$, with $A_i \subseteq AP$.

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Let $\sigma = A_0A_1A_2\dots$, with $A_i \subseteq AP$.

- ▶ $\sigma \models a$ iff $a \in A_0$
- ▶ $\sigma \models \varphi_1 \wedge \varphi_2$ iff $\sigma \models \varphi_1$ and $\sigma \models \varphi_2$
- ▶ $\sigma \models \neg\varphi$ iff $\sigma \not\models \varphi$
- ▶ $\sigma \models \bigcirc\varphi$ iff $A_1A_2\dots \models \varphi$
- ▶ $\sigma \models \varphi \mathbf{U} \psi$ iff
 $\exists j \geq 0$ such that $A_jA_{j+1}\dots \models \psi \wedge \forall 0 \leq i < j, A_iA_{i+1}\dots \models \varphi$

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- ▶ $\sigma \models \Diamond\varphi$ iff $\exists j \geq 0, A_jA_{j+1}\dots \models \varphi$
- ▶ $\sigma \models \Box\varphi$ iff $\forall j \geq 0, A_jA_{j+1}\dots \models \varphi$
- ▶ $\sigma \models \Box\Diamond\varphi$ iff $\forall j \geq 0, \exists i \geq j, A_iA_{i+1}\dots \models \varphi$
- ▶ $\sigma \models \Diamond\Box\varphi$ iff $\exists j \geq 0, \forall i \geq j, A_iA_{i+1}\dots \models \varphi$

If $\sigma = A_0A_1A_2\dots$, $\sigma \models \varphi$ is also written as $\sigma, 0 \models \varphi$. This simply means $A_0A_1A_2\dots \models \varphi$. One can also define $\sigma, i \models \varphi$ to mean $A_iA_{i+1}A_{i+2}\dots \models \varphi$ to talk about a suffix of the word σ satisfying a property.

Transition System Semantics $TS \models \varphi$

Let $TS = (S, S_0, \rightarrow, AP, L)$ be a transition system, and φ an LTL formula over AP

- ▶ For an infinite path fragment π of TS ,

$$\pi \models \varphi \text{ iff } \text{trace}(\pi) \models \varphi$$

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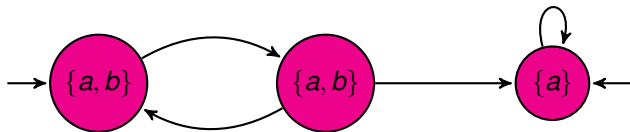
- ▶ $TS \models \varphi$ iff $\text{Traces}(TS) \subseteq L(\varphi)$

Transition System Semantics $TS \models \varphi$

Assume all states in TS are reachable from S_0 .

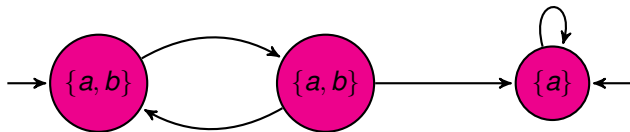
- ▶ $TS \models \varphi$ iff $TS \models L(\varphi)$ iff $Traces(TS) \subseteq L(\varphi)$
- ▶ $TS \models L(\varphi)$ iff $\pi \models \varphi \ \forall \pi \in Paths(TS)$
- ▶ $\pi \models \varphi \ \forall \pi \in Paths(TS)$ iff $s_0 \models \varphi \ \forall s_0 \in S_0$

Example



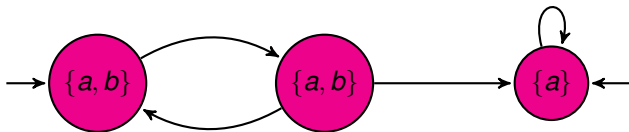
► $TS \models \Box a$,

Example



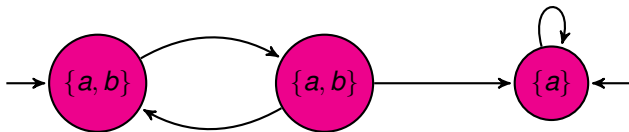
- ▶ $TS \models \Box a$,
- ▶ $TS \not\models \bigcirc(a \wedge b)$

Example



- ▶ $TS \models \Box a,$
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- ▶ $TS \not\models (b \cup (a \wedge \neg b))$

Example



- ▶ $TS \models \Box a$,
- ▶ $TS \not\models \bigcirc(a \wedge b)$
- ▶ $TS \not\models (b \cup (a \wedge \neg b))$
- ▶ $TS \models \Box(\neg b \rightarrow \Box(a \wedge \neg b))$

More Semantics

- ▶ For paths π , $\pi \models \varphi$ iff $\pi \not\models \neg\varphi$

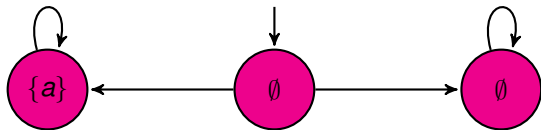
More Semantics

- ▶ For paths π , $\pi \models \varphi$ iff $\pi \not\models \neg\varphi$
 $trace(\pi) \in L(\varphi)$ iff $trace(\pi) \notin L(\neg\varphi) = \overline{L(\varphi)}$
- ▶ $TS \not\models \varphi$ iff $TS \models \neg\varphi$?

More Semantics

- ▶ For paths π , $\pi \models \varphi$ iff $\pi \not\models \neg\varphi$
 $trace(\pi) \in L(\varphi)$ iff $trace(\pi) \notin L(\neg\varphi) = \overline{L(\varphi)}$
- ▶ $TS \not\models \varphi$ iff $TS \models \neg\varphi$?
 - ▶ $TS \models \neg\varphi \rightarrow \forall$ paths π of TS , $\pi \models \neg\varphi$
 - ▶ Thus, $\forall\pi$, $\pi \not\models \varphi$. Hence, $TS \not\models \varphi$
 - ▶ Now assume $TS \not\models \varphi$
 - ▶ Then $\exists$ some path π in TS such that $\pi \models \neg\varphi$
 - ▶ However, there could be another path π' such that $\pi' \models \varphi$
 - ▶ Then $TS \not\models \neg\varphi$ as well
- ▶ Thus, $TS \not\models \varphi \not\equiv TS \models \neg\varphi$.

An Example



$TS \not\models \Diamond a$ and $TS \not\models \Box \neg a$